
From H. R. 9510 to Federal Law: A Narrative Case for Verified Physical AI in the Nation's Oncology Trials

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Abstract.

This review makes a legislative case, for readers who shape medical and artificial intelligence law without robotics or coding experience, that Physical AI Trial Bill H. R. 9510 (v5.0) should be enacted into Federal law. It is built on a single mechanism, verification before generation, in which a software agent proposes a clinical action and a ten-gate verification, validation, and uncertainty quantification examination either accepts it, escalates it to a qualified human, or blocks it before it can reach a patient. The argument is organized as eight emotional pillars that legislative-advocacy research finds most persuasive: compassion, fear of preventable harm, moral outrage, hope, responsibility, protection of vulnerable people, trust, and urgency. Each pillar pairs a human appeal with a credible, cited fact, so the story is memorable and the evidence is actionable. The review draws on a documented engineering lineage, including a surgical-humanoid assurance run that passed 172 of 172 automated tests across a ten-gate suite, and on the published advocacy literature describing how testimony, coalition building, and policy entrepreneurship move a bill through markup and reconciliation. Throughout, three quiet questions recur: how did we ever ask humans alone to carry this task, why did we trust a clinical-trial system with so many chances for compounding human error, and why would human peer review remain the only check on increasingly complex and voluminous artificial intelligence outputs. The conclusion is a single call to action with a specific instrument attached.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords Physical AI | Oncology clinical trials | Verification before generation | VVUQ | Medical AI legislation | H. R. 9510 | Patient safety | Emotional narrative advocacy

1 Introduction: a quiet question in the committee room

This review is written for a reader who shapes medical and artificial intelligence law but who has never driven a surgical robot or written a line of verification code. It therefore defines its terms as it goes, and it

asks one quiet question in eight different ways: how did we ever ask human beings, alone, to carry a task this large without help?

A few plain definitions make the rest of the story readable. *Physical AI* is artificial intelligence that acts in the physical world through a machine, such as a surgical humanoid in an oncology trial, rather than only returning text on a screen. *Verification before generation* is the principle at the center of H. R. 9510 [1]: a machine may propose an action, but the proposal is checked against fixed, published standards before anything reaches a patient. *Claude Code* is the software agent that drafts the candidate procedure, *Codex* is a second, independent agent that reviews it, and *VVUQ*, meaning verification, validation, and uncertainty quantification, is the ten-gate examination the proposal must pass. The examination resolves in exactly one of three ways: ACCEPT and execute under audit, ESCALATE to a qualified human, or BLOCK before execution. Figure 1 shows the whole mechanism at a glance.

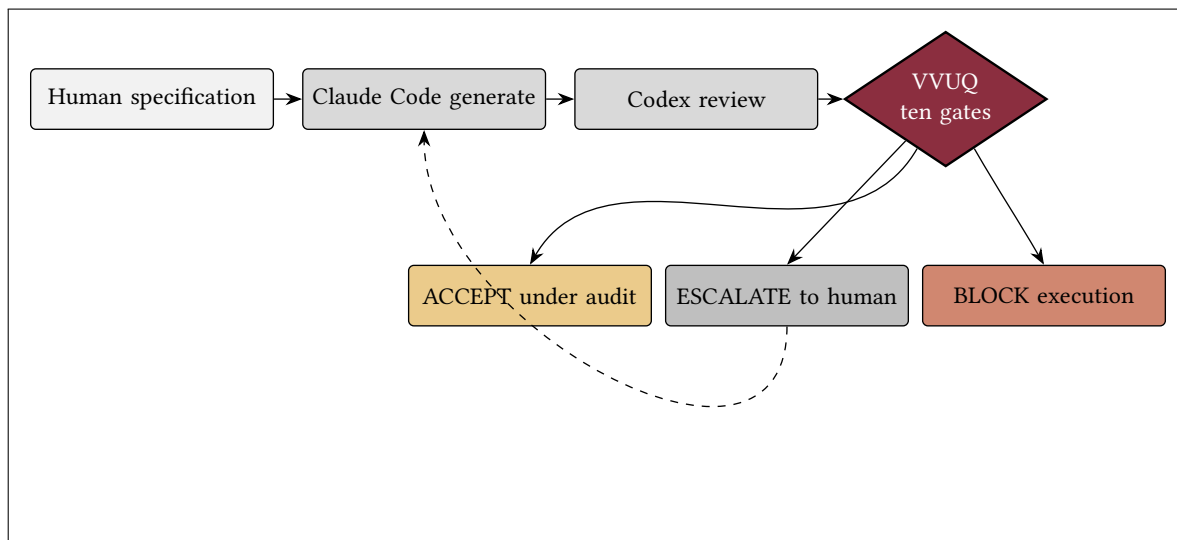


Figure 1. Verification before generation (Mermaid 01 reproduced as colored TikZ). A human specification is drafted by one agent, reviewed by a second, and admitted only after ten gates resolve to ACCEPT, ESCALATE, or BLOCK.

The three guiding questions return throughout. How did we ever utilize humans alone for this challenging task? Why did we ever fully trust the prior human clinical-trial system if there were so many opportunities for compounding human error? And why would the human peer-review process continue to be used for increasingly complex and voluminous artificial intelligence outputs? These are asked politely, not to diminish the clinicians who have carried the system on their shoulders, but to notice that the load has outgrown unaided human attention.

The advocacy literature is clear that emotion and evidence must travel together. Legislators are moved by compelling, concise stories that make an abstract technology tangible [2], yet they weigh the credibility of the presenter and the underlying data at least as heavily as the intensity of the appeal [3]. This review therefore organizes its case as eight emotional pillars, and pairs each with a cited fact. Figure 2 previews that path, from the most common appeal, compassion, to the closing appeal, urgency, all converging on a single action.

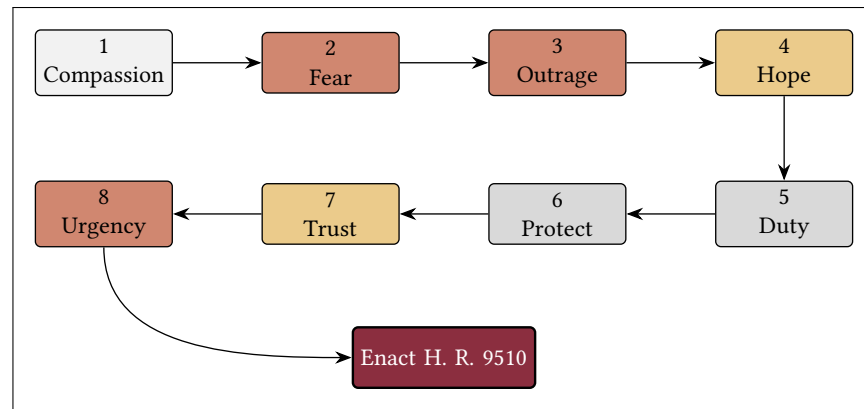


Figure 2. The eight emotional pillars (Mermaid 02 reproduced as colored TikZ). The argument accumulates from compassion to urgency and converges on a single legislative request.

The remainder of the review takes the pillars in turn. Each section opens with a human appeal, anchors it in credible evidence, defines any term on first use, and closes by naming the specific provision of H. R. 9510 that answers it.

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2 Compassion and empathy: what happens if we do nothing?

Every successful health law begins where medicine itself begins, with a person. A committee may hear statistics from experts all day, but a patient who describes a long wait for a trial slot, or a parent who describes a child with a rare cancer, creates an emotional anchor that the numbers alone cannot. Patient stories do more than move a room: they supply contextual depth that helps a decision maker grasp the real-world stakes of a bill [2]. The honest question such a story poses is the one a committee cannot avoid. What happens if we do nothing?

Consider the journey the author has documented elsewhere, of a patient moving through a regulated Physical AI oncology trial from referral to follow-up [4]. At each step the patient is asked to trust a system. Figure 3 traces that path. Compassion asks us to see the person inside it, and to remember that for many patients the alternative to a verified, well-run trial is not a safer status quo but no option at all.

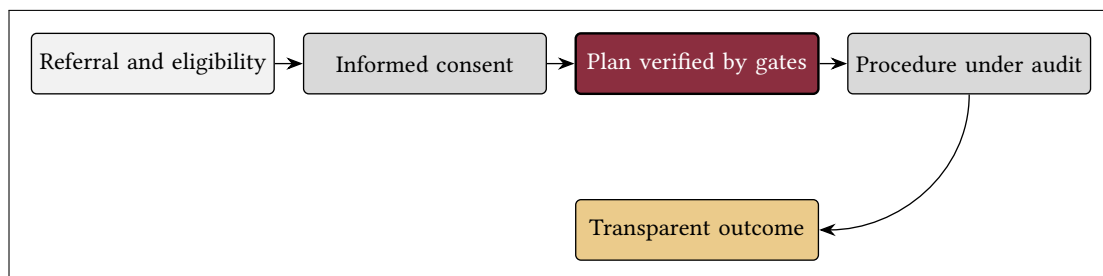


Figure 3. A patient's journey through a verified trial (Mermaid 11 as colored TikZ): consent and verification precede any action, and the outcome is recorded transparently.

The four faces of this appeal are familiar from rare-disease legislation, cancer screening laws, expanded coverage, and right-to-try initiatives. They are summarized in Table 1, each paired with the cost of inaction that a patient would name in testimony.

Who	What they face	What “doing nothing” means
Patient with serious illness	Few trials, long waits, narrow eligibility	Time lost that cannot be recovered
Family caregiver	Coordinating care across a fragmented system	Burden carried alone, errors unseen
Child with a rare cancer	Too few cases for ordinary evidence	A treatable disease left unstudied
Person without access	Distance, cost, or coverage barriers	Care that exists but never arrives

Table 1: Four patient archetypes and the human cost of inaction.

The point is not to substitute feeling for evidence. It is to make the evidence memorable by attaching it to the people it protects. H. R. 9510 answers the compassion appeal not with a slogan but with structure: a verified pathway that a patient can actually enter, with consent confirmed and every action recorded. The quiet question follows naturally. If a careful, checked system can be offered, how did we ever ask patients to accept less?

3 Fear of preventable harm: the errors we can no longer accept

The second appeal is the fear of harm that did not have to happen. A generation of health policy, from drug-safety regulation to device oversight to electronic health record requirements, was built on stories of missed diagnoses, medication errors, unsafe products, and delayed treatment. The fear is not abstract. The landmark report *To Err Is Human* estimated that tens of thousands of Americans die each year from preventable medical error [5], and a later analysis ranked medical error among the leading causes of death in the United States [6]. The underlying message of this pillar is plain: without a better check, more people will be harmed than need to be.

Why does error accumulate? Because the legacy clinical-trial and peer-review system passes work through many manual handoffs, and each handoff is an independent opportunity for a mistake that compounds downstream. The verified path inserts an automated gate at each step, so a defect is caught where it occurs rather than discovered after a patient is affected. Figure 4 draws the contrast, and the engineering record behind it is concrete: a surgical-humanoid assurance run passed 172 of 172 automated tests, 64 of them inside a ten-gate suite, before any action was permitted [7].

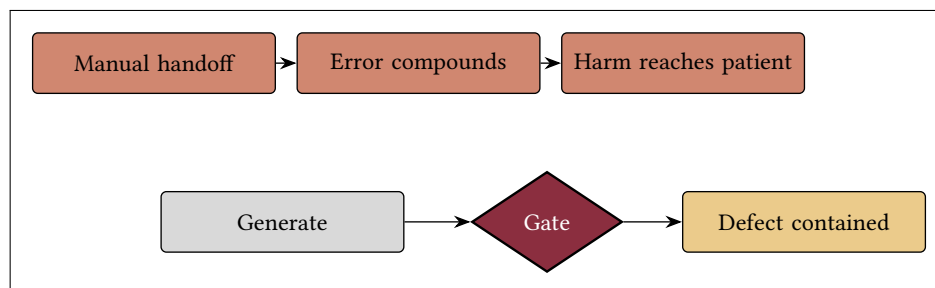


Figure 4. Compounding error, above, versus a contained defect, below (Mermaid 04 and 20 as colored TikZ). The gate stops a defect at its source.

Table 2 sets the two paths side by side. The comparison is the kind of scenario-based risk information that helps a committee visualize a future loss and turn a what-if into a clear policy goal.

Step	Legacy path	Verified path
Where a defect is caught	After the fact, if at all	At the gate where it occurs
Who must notice it	A tired human, unaided	An automated check, every time
Record of the decision	Often informal	Hash-chained and auditable
Effect of one error	Compounds downstream	Contained and escalated

Table 2: The legacy path and the verified path, step by step.

A bill cannot promise that nothing will ever go wrong. It can promise that the system checks itself at every step, so the same mistake is not made twice and is never made invisibly. When committee *markup* sharpens this language, the safety guarantee becomes part of the permanent record. The quiet question returns: why did we ever fully trust a process with so many chances for compounding human error?

4 Moral outrage: this should not happen in America

Many health laws gain momentum after a perceived injustice: an insurance denial, an unaffordable treatment, a discriminatory algorithm, or plainly unequal access to care. The framing becomes a sentence a member can repeat on the floor. This is unfair, and it should not happen here. Outrage is effective precisely because it draws both public attention and press coverage to a specific, fixable wrong, which is why coalition building around a documented injustice is such a powerful legislative move.

The injustice here is documented, not asserted. A widely used clinical risk algorithm was shown to systematically underestimate the needs of Black patients, because it used past spending as a proxy for illness, and spending itself reflected unequal access [8]. An unverified algorithm can encode a disparity and then scale it across every patient it touches. The author’s comparison of proposed bills places patient priority, rather than institutional convenience, at the center of the remedy [9]. Figure 5 shows how a documented gap becomes an audited safeguard.

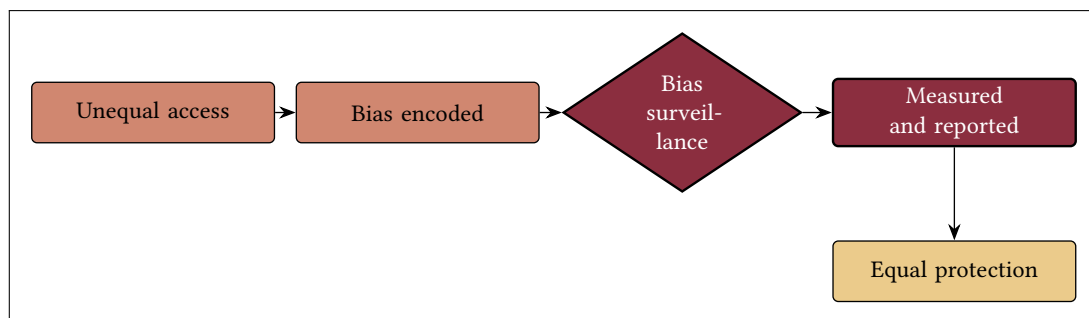


Figure 5. From a documented disparity to an audited safeguard (Mermaid 15 as colored TikZ). Surveillance turns an unfair gap into a measured, correctable number.

Table 3 contrasts the unverified status quo with the remedy the bill provides, so the outrage points somewhere constructive.

Concern	Unverified status quo	Remedy in H. R. 9510
Algorithmic bias	Hidden, scaled silently	Subgroup surveillance, reported
Unequal access	Tolerated as background	Access recorded and audited
Accountability	Diffuse, hard to trace	Named record, public report
Correction	Slow or never	Triggered by a measured gap

Table 3: Turning a documented injustice into a statutory check.

The remedy is not outrage for its own sake. It is a precise statutory check that converts a documented disparity into a measured, reported, and correctable number. A committee that has seen the data can ask the right question. If we can measure the unfairness, why would we choose not to?

5 Hope: the opportunity to save lives if we act

Not every successful health law is built on fear. The most durable ones also offer hope: earlier cancer detection, reduced physician burnout, better rural access, and faster discovery. The convergence of human and machine intelligence in medicine is already improving diagnosis and freeing clinician time for the work only people can do [10]. For medical AI, the hopeful claim is concrete and, importantly, it is grounded. We have an opportunity to save lives if we act, and the tools to do so responsibly have already been built and tested.

That grounding matters, because hope without evidence is only a wish. The author’s work traces a documented lineage, from accelerated patient prediction across extensive simulations [11] to in-silico trials that lower the cost and time of compliance [12]. This is the kind of policy entrepreneurship that pairs a clearly identified problem with a specific, demonstrated solution. Figure 6 shows the lineage on which the hope rests.

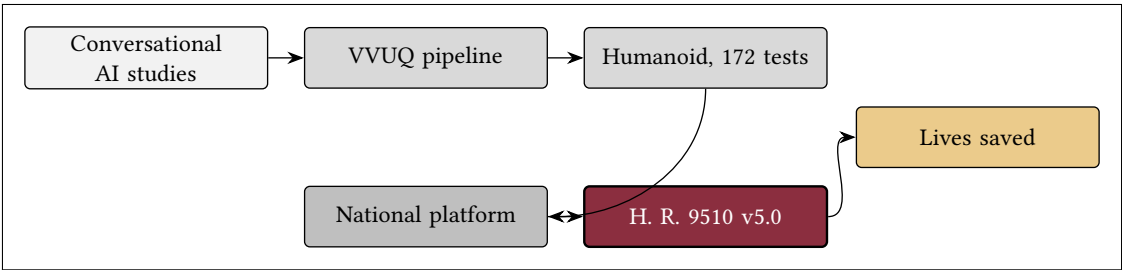


Figure 6. Hope grounded in a documented lineage (Mermaid 06 as colored TikZ): each step is a real deposit on which the next was built.

Table 4 names four near-term benefits and the bill mechanism that enables each, so the optimism is specific rather than vague.

Near-term benefit	Mechanism the bill enables
Earlier cancer detection	Verified models cleared for trial use
Reduced physician burnout	Routine checks automated, audited
Improved rural access	Standardized, portable trial platform
Faster, cheaper discovery	In-silico stages validated before live work

Table 4: Four hopeful outcomes and the provisions that enable them.

Hope, paired with evidence, gives a legislator something to vote for rather than only something to vote against. The quiet question turns gentle here. If the tools already exist and have passed their tests, why would we wait to let patients benefit from them?

6 Responsibility and duty: you have the power to prevent this

The fifth appeal is addressed to the legislator directly. It is not sympathy but obligation. You have the power to prevent this, and future generations will judge what we do now. This is the moment the narrative turns from describing a problem to naming the body that can solve it and the precise instrument, H. R. 9510, that lets it. Duty creates a sense of agency that sympathy alone does not.

Responsibility is exercised through a known sequence of actions. *Advocacy* speaks in favor of safe medical AI. *Coalition building* assembles patient groups, hospital networks, technology developers, and medical boards so the bill visibly carries wide support. *Policy entrepreneurship* matches the documented problem of unverified AI to this specific legislative fix. And when two chambers pass different texts, *reconciliation* resolves them into one. Credible *testimony at committee hearings* does not always change a vote, but it reliably informs the record, shapes the floor debate, and clarifies the language a committee writes during *markup* [3]. Figure 7 shows the path on which these actions operate.

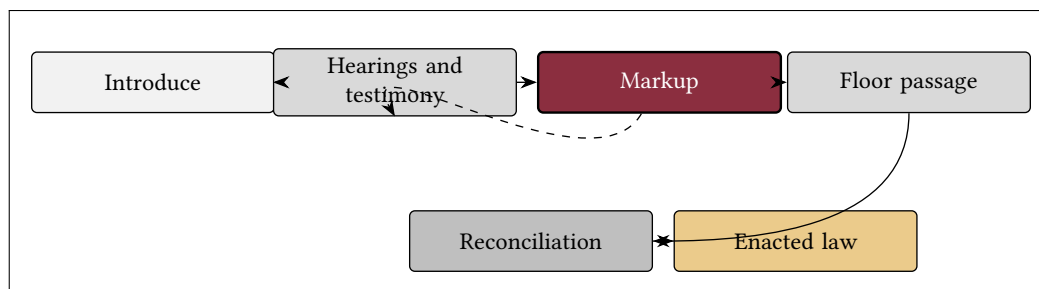


Figure 7. The duty to move a bill from introduction to enactment (Mermaid 03 and 14 as colored TikZ), with testimony feeding markup and reconciliation joining two chambers.

Table 5 maps each main action to the committee step where it has the most effect, so a member can see exactly where to apply effort.

Main action	Where it has effect
Advocacy	Building awareness before referral
Coalition building	Demonstrating support at hearings
Policy entrepreneurship	Framing the problem during markup
Reconciliation	Resolving House and Senate texts

Table 5: Each main action and the legislative step it serves.

Responsibility is the bridge from feeling to action. It tells the committee that the next move is theirs and that the record will remember it. The quiet question becomes a charge. If the power to prevent foreseeable harm rests here, why would we leave it unused?

7 Protection of vulnerable people: a duty of care, written down

The sixth appeal is protective rather than purely compassionate. Health debates return again and again to the groups most exposed to harm: children, the elderly, disabled patients, veterans, and rural communities. For medical AI the specific worries are bias against minority populations, errors that fall hardest on the vulnerable, and consent that is assumed rather than confirmed. The same bias that the previous section measured falls most heavily on the people least able to absorb it [8], which is why a national action plan for AI as a medical device places real-world performance monitoring across subgroups at its center [13].

Protection is strongest when it is built into the procedure rather than promised afterward. The author's site-documentation and consent work, including an adaption of 21 CFR Part 50 for end-to-end Physical AI trials, makes informed consent a verified, logged step rather than a signature gathered in haste [14, 15]. Figure 8 shows the three gates a subgroup passes before its results are accepted.

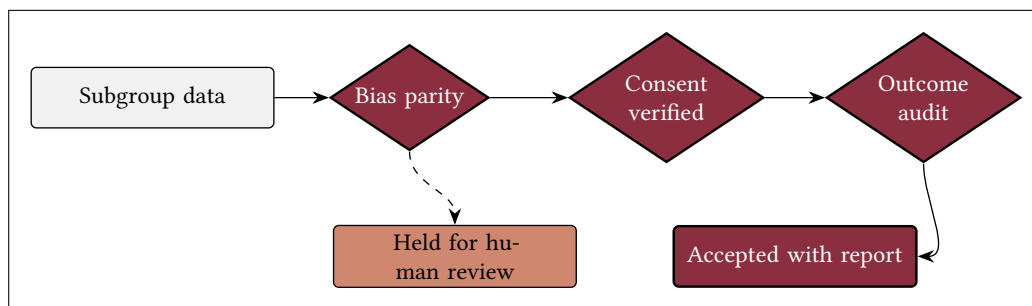


Figure 8. Safeguards written into the procedure (Mermaid 15 as colored TikZ): parity, consent, and audit each guard the accept, or hold the work for a human.

Table 6 lists each protected group, its specific risk, and the safeguard that answers it.

Group	Specific risk	Safeguard
Children	Too few cases for ordinary evidence	Subgroup audit and reporting
Elderly and disabled	Models trained on others	Bias-parity surveillance
Veterans	Fragmented records	Verified, auditable record
Rural communities	Distance and access gaps	Portable, standardized platform

Table 6: Protected groups, their risks, and the bill safeguards.

Protection that is written into statute, measured, and reported is stronger than protection that depends on the good intentions of whoever happens to be on duty. The quiet question is gentle but firm. If we can guarantee the check in law, why would we leave the most vulnerable patients relying on hope alone?

8 Trust and reassurance: why this technology can be trusted

Medical AI bills face honest public skepticism, so the seventh appeal is reassurance. Trust is not asserted; it is built from named supports: physician endorsement, academic review, mandatory safety testing, transparency provisions, pilot results, and an audit trail that anyone can inspect. The objective is a committee that can tell constituents, with evidence, that this technology can be trusted. Figure 9 names the supports that hold that trust up.

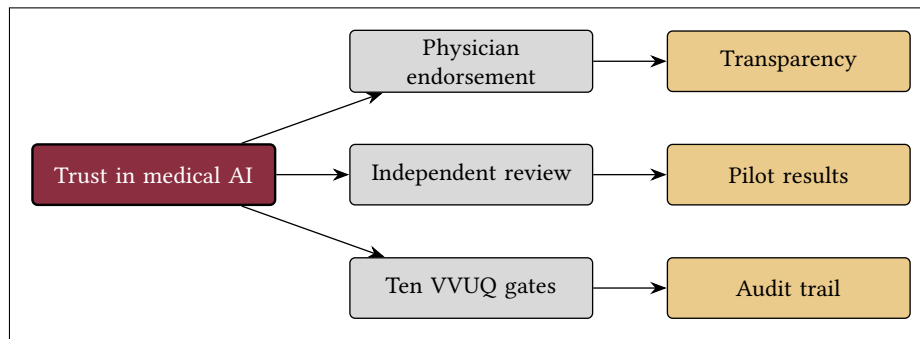


Figure 9. Trust scaffolding (Mermaid 10 as colored TikZ): confidence rests on named, inspectable supports, not on assurance alone.

The strongest support is the most concrete. Each of the ten gates is bound to a published consensus standard already used in real engineering practice, such as ASME V&V 40-2018 for assessing the credibility of computational models in medical devices [16], and the surgical-humanoid run cleared all of them, passing 172 of 172 automated tests [7, 17]. Figure 10 shows the funnel, and Table 7 binds representative gates to their standards and thresholds, which is what makes the assurance argument traceable to a regulator.

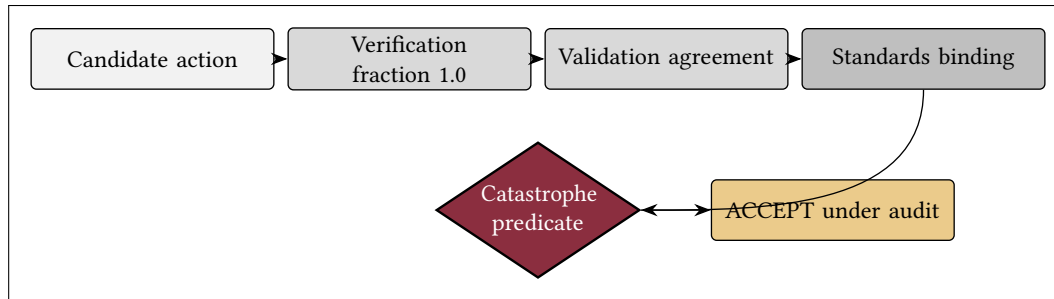


Figure 10. The ten-gate funnel (Mermaid 07 as colored TikZ): each gate is a published threshold, and only a clean pass reaches ACCEPT.

Gate	Bound standard or basis	Pass threshold
Verification fraction	ASME V&V 40-2018 credibility	exactly 1.0
Validation agreement	Reference cohort comparison	up to 1.00
Relative error	Numerical tolerance	as tight as 0.01
Catastrophe predicate	Hard safety rule	block before execution

Table 7: Representative gates bound to standards and thresholds.

Reassurance works when it is specific. A named standard and a published test result persuade where a promise of safety cannot. The quiet question follows. If every check can be bound to a standard and shown to pass, why would human review alone remain the only gate on increasingly complex AI outputs?

9 Urgency: patients are waiting now

The final appeal is timing. Legislation accelerates when delay is shown to have a cost: patients are waiting now, lives are being lost now, and the technology is advancing faster than the rules that govern it. But urgency must be handled with care. Research on fear appeals finds that they motivate action only when the audience also believes a solution is within reach; a catastrophe presented without a credible remedy tends to paralyze rather than move [18]. So this section pairs the pressure of time with a remedy already in hand.

The gap is real. Capability, in the form of Physical AI, surgical humanoids, and autonomous pipelines, is rising quickly, while regulation, built for an earlier era and supplemented mainly by guidance, lags behind [13]. The distance between the two curves is where preventable harm accumulates. Figure 11 shows the gap and the single instrument that closes it.

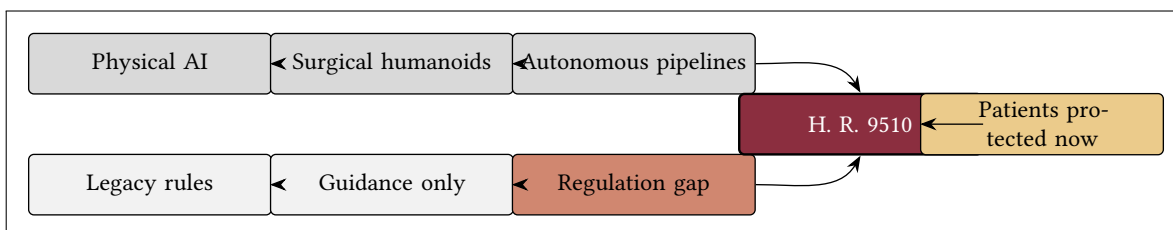


Figure 11. Closing the capability-regulation gap (Mermaid 16 and 21 as colored TikZ): the Act is the bridge that joins fast-moving capability to lagging rules.

Table 8 sets the cost of another year of delay against the cost of acting, so the urgency points toward a decision rather than toward despair.

Dimension	Effect of one more year
Patients waiting	Trial access delayed for those with the least time
Preventable harm	Unverified tools deployed without a gate
Public trust	Skepticism hardens without a transparent record
Cost of acting	Bounded, authorized, and budget-scored in the bill

Table 8: The cost of delay against the bounded cost of acting.

The eight appeals converge on one request. They are not eight separate arguments but eight doors into the same room, and each one opens onto a specific provision of the bill, summarized in Table 9. This is the bridge the advocacy literature describes, where the emotional story makes the data memorable and the data makes the story actionable [3, 18].

Pillar	Provision of H. R. 9510 that answers it
Compassion	A verified pathway a patient can enter, with consent confirmed and recorded
Fear of harm	The mandatory ten-gate verification before any action executes
Moral outrage	Subgroup bias surveillance, measured and publicly reported
Hope	The authorization that funds a tested national trial platform
Responsibility	The conforming amendments that make verification the standard of care
Protection	Verified informed consent and a per-subgroup outcome audit
Trust	Each gate bound to a published standard, with a hash-chained audit trail
Urgency	A defined effective date and a budget-scored, bounded cost of acting

Table 9: The eight pillars and the specific provisions of H. R. 9510 that answer them.

We therefore ask the committee to enact H. R. 9510 (v5.0) into Federal law [1]. The quiet questions, asked eight times, resolve into a single answer: we no longer have to carry this task with unaided human attention, and we should not ask patients to wait while we decide to stop.

10 Appendix A: figure compendium

This appendix reproduces the remaining figures from the colored Mermaid catalog (review/mermaid/) as colored TikZ, so the manuscript carries the full set of exhibits in the strict palette. Each caption names the Mermaid source.

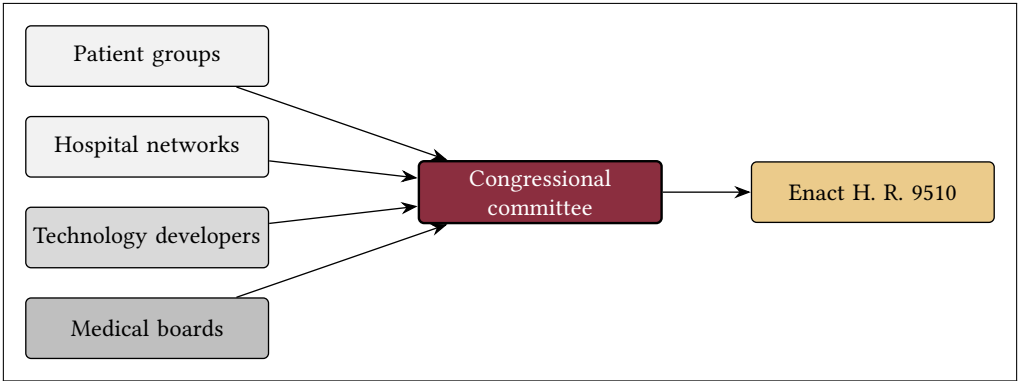


Figure 12. The coalition map (Mermaid 05): four constituencies converge on one committee.

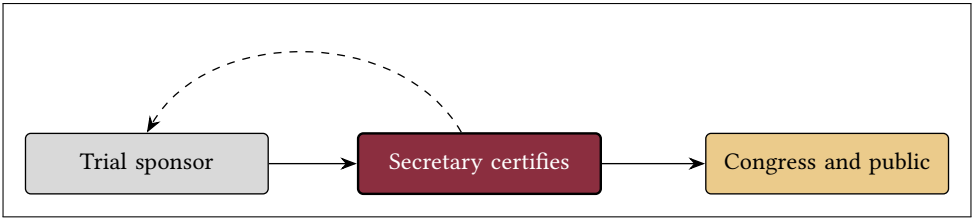


Figure 13. The annual reporting sequence (Mermaid 08): a verified record is certified, then reported, with a correction loop.

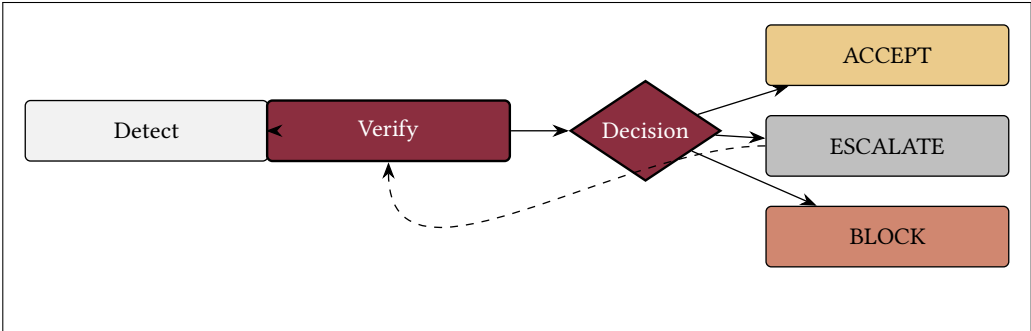


Figure 14. The verification decision state machine (Mermaid 09): three guarded outcomes, with escalation returning for revision.

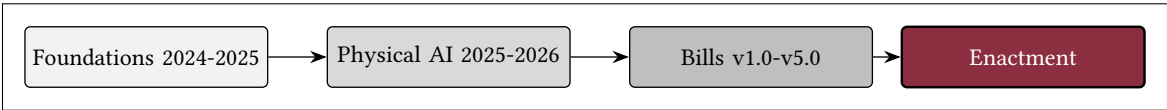


Figure 15. The evidence-and-legislation timeline (Mermaid 12): four phases from research to enactment.

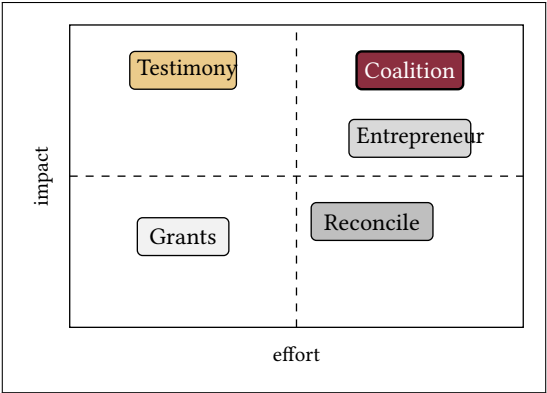


Figure 16. Legislative actions by effort and impact (Mermaid 13): high-impact actions sit in the upper half.

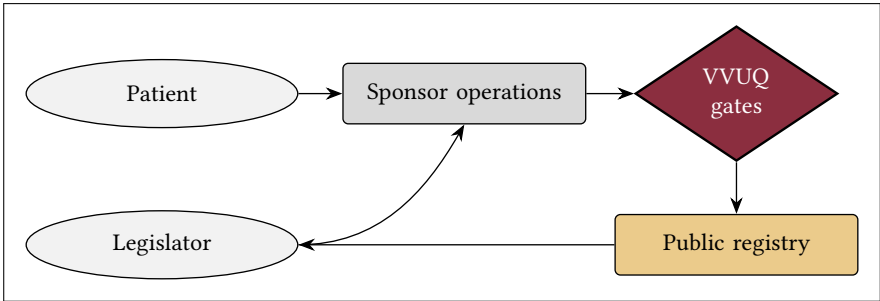


Figure 17. System context (Mermaid 17): people and systems and how they relate around the verified platform.

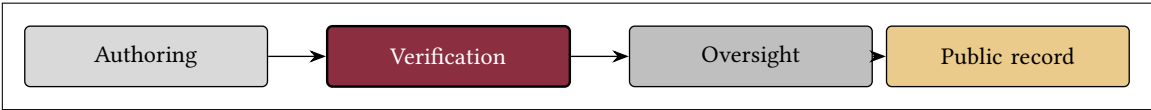


Figure 18. The composite oversight state machine (Mermaid 18): three super-states lead to a public record.

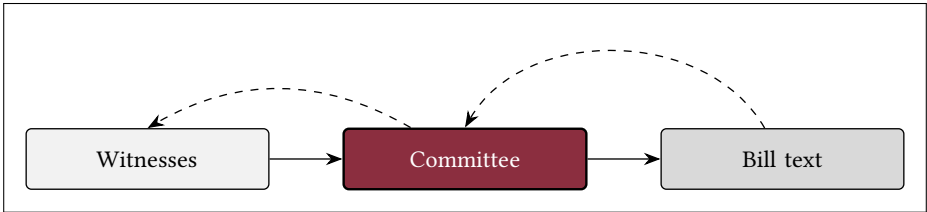


Figure 19. Markup and testimony (Mermaid 19): testimony and amendment loops build the public record.

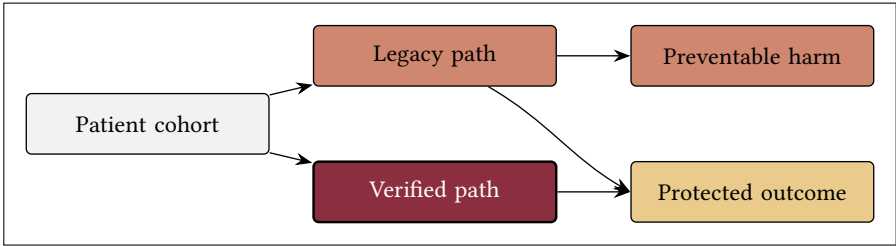


Figure 20. Patient throughput (Mermaid 20): the verified path moves more of the cohort to a protected outcome.

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Ethical Disclosures

The author declares no competing interests. This is an independent narrative review and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. H. R. 9510 is an illustrative bill number; it is not enacted law and this review is not legal advice. All supporting numbers are simulation or illustrative results unless tied to a cited source.

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Data Availability

All sources, figures, and prompt records supporting this review are openly available in the `kevinkawchak/law-physical-ai-trials` repository under `review/` (the prompt, the four sub-prompts, the colored Mermaid catalog, and the three narrative stages), and draw on the public records in `kevinkawchak/cancer-automated` and the author's Zenodo deposits. All data sets are synthetic or illustrative; no patient-identifiable information is included.